# Huawei Access Node JAY-ACC-39 Optical Transport Failure SOP (ALARM_6323)

**Node:** JAY-ACC-39 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6323

## Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for addressing Optical Transport Failures on the Huawei Access Node JAY-ACC-39, specifically related to SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. The failure domain is attributed to the Optical Transport Network (OTN) layer, which is responsible for transmitting client signals over optical fibers. The OTN layer employs the Generic Framing Procedure (GFP) protocol to encapsulate client signals, which are then transmitted over the optical transport network. The Huawei Access Node JAY-ACC-39, being a Huawei Access node, employs the Huawei's proprietary OTN technology, which uses the Enhanced Generic Framing Procedure (EGFP) protocol to provide enhanced framing and multiplexing capabilities. The Optical Transport Failure (ALARM_6323) is triggered when the receiver optical power (Rx) or transmitter optical power (Tx) falls below a certain threshold, indicating a potential issue with the SFP/XFP module, fiber cut, or other optical transport-related problems.

## Pre-requisite Checks

- Step: Verify Redundant Link Status, Description: Ensure that the redundant link is up and functioning correctly to prevent network downtime during the troubleshooting process.

- Step: Verify Physical Connection, Description: Inspect the physical connections between the Huawei Access Node JAY-ACC-39 and the neighboring nodes to ensure that all connections are secure and not damaged.

- Step: Verify Environmental Conditions, Description: Check the environmental conditions, such as temperature, humidity, and vibration, to ensure that they are within the recommended operating range for the Huawei Access Node JAY-ACC-39.

## Potential Root Causes

| Cause | Probability |
|---|---|
| SFP/XFP degradation | High |
| Fiber cut | Medium |
| Rx/Tx power loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display optical link interface, Expected Output: Optical Link Interface Information, Description: This command displays the optical link interface information, including the receive and transmit optical power levels, which can help identify potential issues with the SFP/XFP module or fiber cut.
```

```
Command: display optical transmit power, Expected Output: Optical Transmit Power Information, Description: This command displays the optical transmit power information, which can help identify potential issues with the Tx power levels.

Command: display optical receive power, Expected Output: Optical Receive Power Information, Description: This command displays the optical receive power information, which can help identify potential issues with the Rx power levels.
```

## Resolution Steps

Step 1: Step: Replace SFP/XFP Module, Command: replace optical module, Description: Replace the SFP/XFP module with a new one to resolve the issue. Ensure that the new module is compatible with the Huawei Access Node JAY-ACC-39 and is properly seated and secured.

Step 2: Step: Repair or Replace Fiber, Command: repair or replace fiber, Description: Repair or replace the fiber cable to resolve the issue. Ensure that the new fiber cable is compatible with the Huawei Access Node JAY-ACC-39 and is properly connected and secured.

Step 3: Step: Adjust Optical Power Levels, Command: adjust optical power level, Description: Adjust the optical power levels to resolve the issue. Ensure that the adjusted power levels are within the recommended operating range for the Huawei Access Node JAY-ACC-39.

## Post-Check Verification

```
Command: display optical link interface, Expected Output: Optical Link Interface Information, Description: Verify that the optical link interface information displays the correct receive and transmit optical power levels.

Command: display optical transmit power, Expected Output: Optical Transmit Power Information, Description: Verify that the optical transmit power information displays the correct power levels.

Command: display optical receive power, Expected Output: Optical Receive Power Information, Description: Verify that the optical receive power information displays the correct power levels.
```

## Roll-back Procedures

- Step: Restore Original SFP/XFP Module, Command: restore original optical module, Description: Restore the original SFP/XFP module to resolve the issue. Ensure that the original module is properly seated and secured.

- Step: Restore Original Fiber, Command: restore original fiber, Description: Restore the original fiber cable to resolve the issue. Ensure that the original fiber cable is properly connected and secured.

- Step: Restore Original Optical Power Levels, Command: restore original optical power level, Description: Restore the original optical power levels to resolve the issue. Ensure that the restored power levels are within the recommended operating range for the Huawei Access Node JAY-ACC-39.